24/04/2026

In [1]:
```
import pandas as pd
```

In [2]:
```
df=pd.read_csv("cleaned_GlobalWeatherRepository.csv")
```

DATA VISUALIZATION

In [3]:
```
import matplotlib.pyplot as plt
import seaborn as sns
```

In [4]:
```
df
```

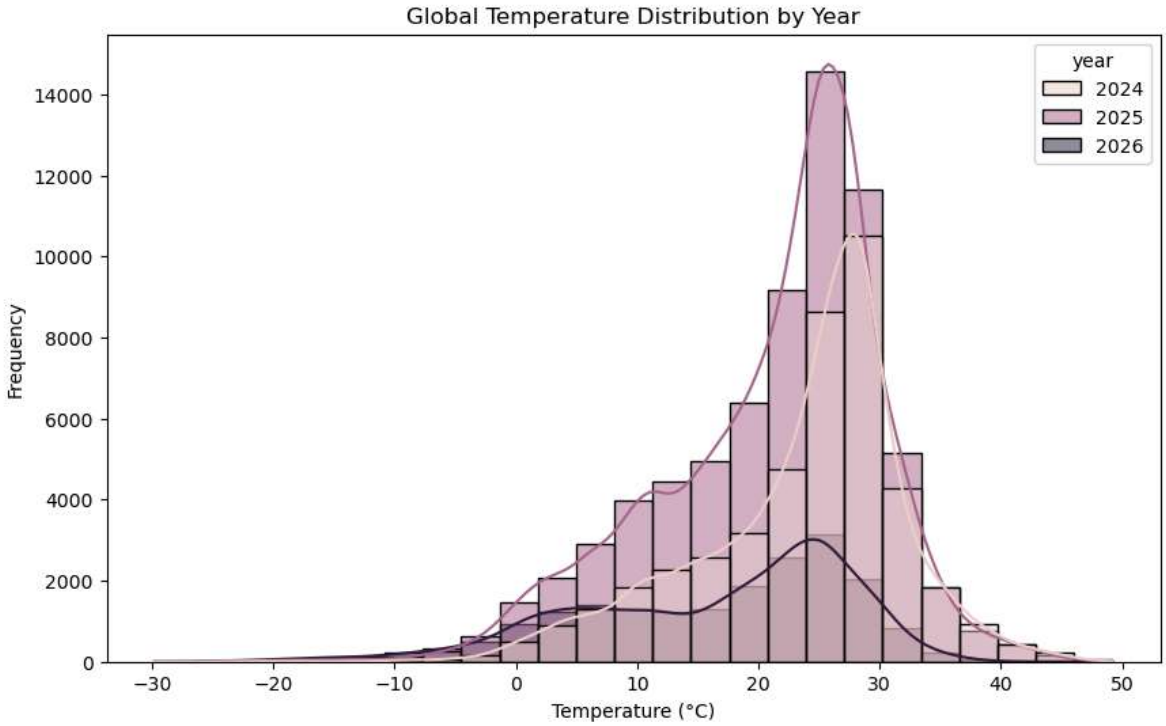
Out[4]:

	country	location_name	latitude	longitude	timezone	last_updated
0	Afghanistan	Kabul	34.5200	69.1800	Asia/Kabul	2024-05-16 13:15:00
1	Albania	Tirana	41.3300	19.8200	Europe/Tirane	2024-05-16 10:45:00
2	Algeria	Algiers	36.7600	3.0500	Africa/Algiers	2024-05-16 09:45:00
3	Andorra	Andorra La Vella	42.5000	1.5200	Europe/Andorra	2024-05-16 10:45:00
4	Angola	Luanda	-8.8400	13.2300	Africa/Luanda	2024-05-16 09:45:00
...	...	...	...	...	...	...
135057	Venezuela	Caracas	10.5000	-66.9167	America/Caracas	2026-04-12 02:30:00
135058	Vietnam	Hanoi	21.0333	105.8500	Asia/Bangkok	2026-04-12 13:30:00
135059	Yemen	Sanaa	15.3547	44.2067	Asia/Aden	2026-04-12 09:30:00
135060	Zambia	Lusaka	-15.4167	28.2833	Africa/Lusaka	2026-04-12 08:30:00
135061	Zimbabwe	Harare	-17.8178	31.0447	Africa/Harare	2026-04-12 08:30:00

135062 rows × 32 columns

Global Temperature Distribution

In [5]:
```
df.columns = df.columns.str.strip()

plt.figure(figsize=(10,6))

sns.histplot(
    data=df,
    x="temperature_celsius",
    hue="year",
    bins=25,
    kde=True
)

plt.xlabel('Temperature (°C)')
plt.ylabel('Frequency')
plt.title('Global Temperature Distribution by Year')

plt.show()
```

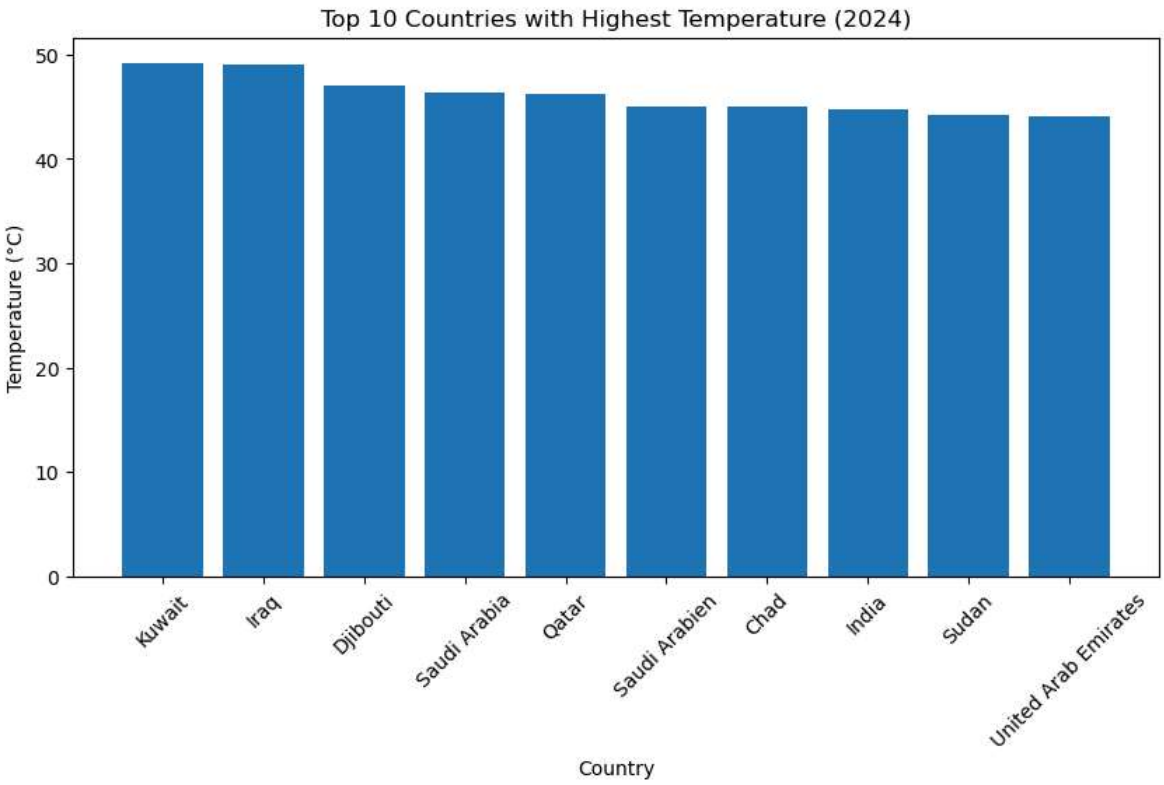


- The graph shows how global temperature distribution varies across different years, helping to compare changes and identify overall trends.

High temperature of top 10 contries

In [6]:
```
df_2024 = df[df["year"] == 2024]
country_temp = df_2024.groupby('country')['temperature_celsius'].max()
top10 = country_temp.sort_values(ascending=False).head(10)
plt.figure(figsize=(10,5))
plt.bar(top10.index, top10.values)

plt.xticks(rotation=45)
plt.xlabel('Country')
plt.ylabel('Temperature (°C)')
plt.title('Top 10 Countries with Highest Temperature (2024)')
```

```
plt.show()
```

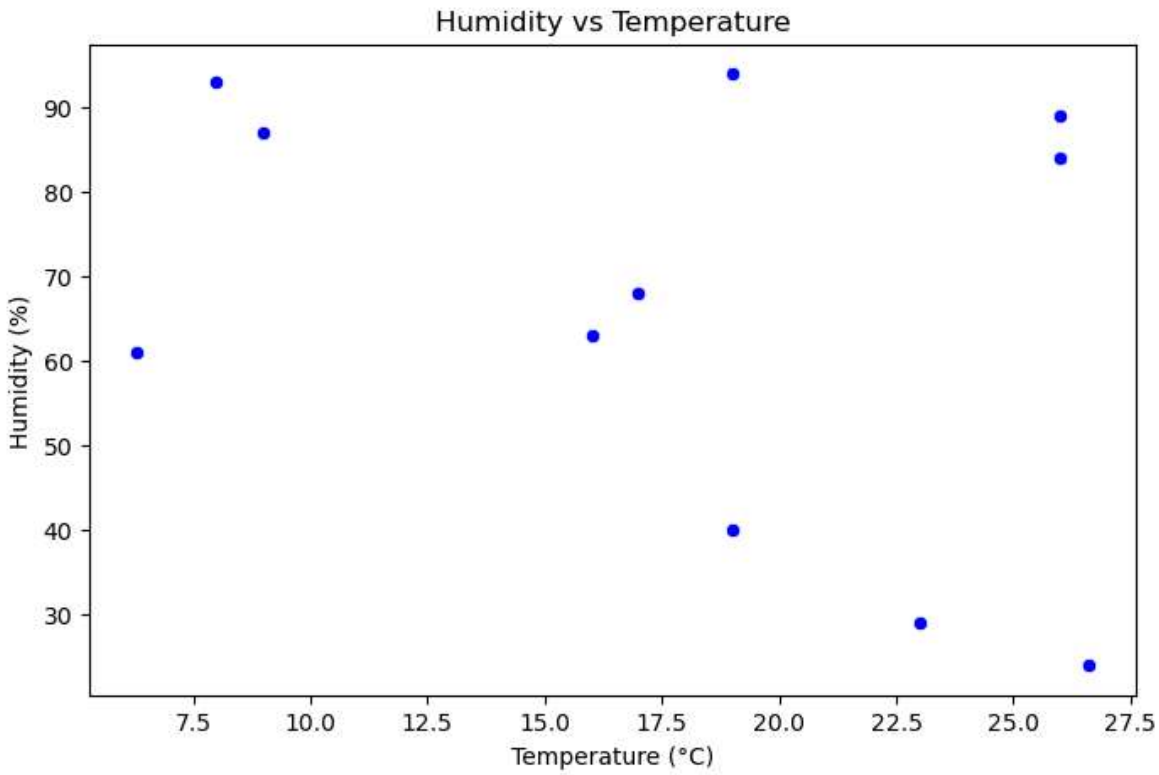


- The chart shows the top 10 hottest countries with temperature ranging about 45°C to 50°C.
- All countries have very high temperature ,with only small diffrence between them.

Temperature vs Humidity

In [7]:
```
gf=df.loc[:10]
```

In [8]:
```
plt.figure(figsize=(8,5))

sns.scatterplot(x=gf["temperature_celsius"], y=gf["humidity"], color="Blue")

plt.title("Humidity vs Temperature")
plt.xlabel("Temperature (°C)")
plt.ylabel("Humidity (%)")

plt.show()
```

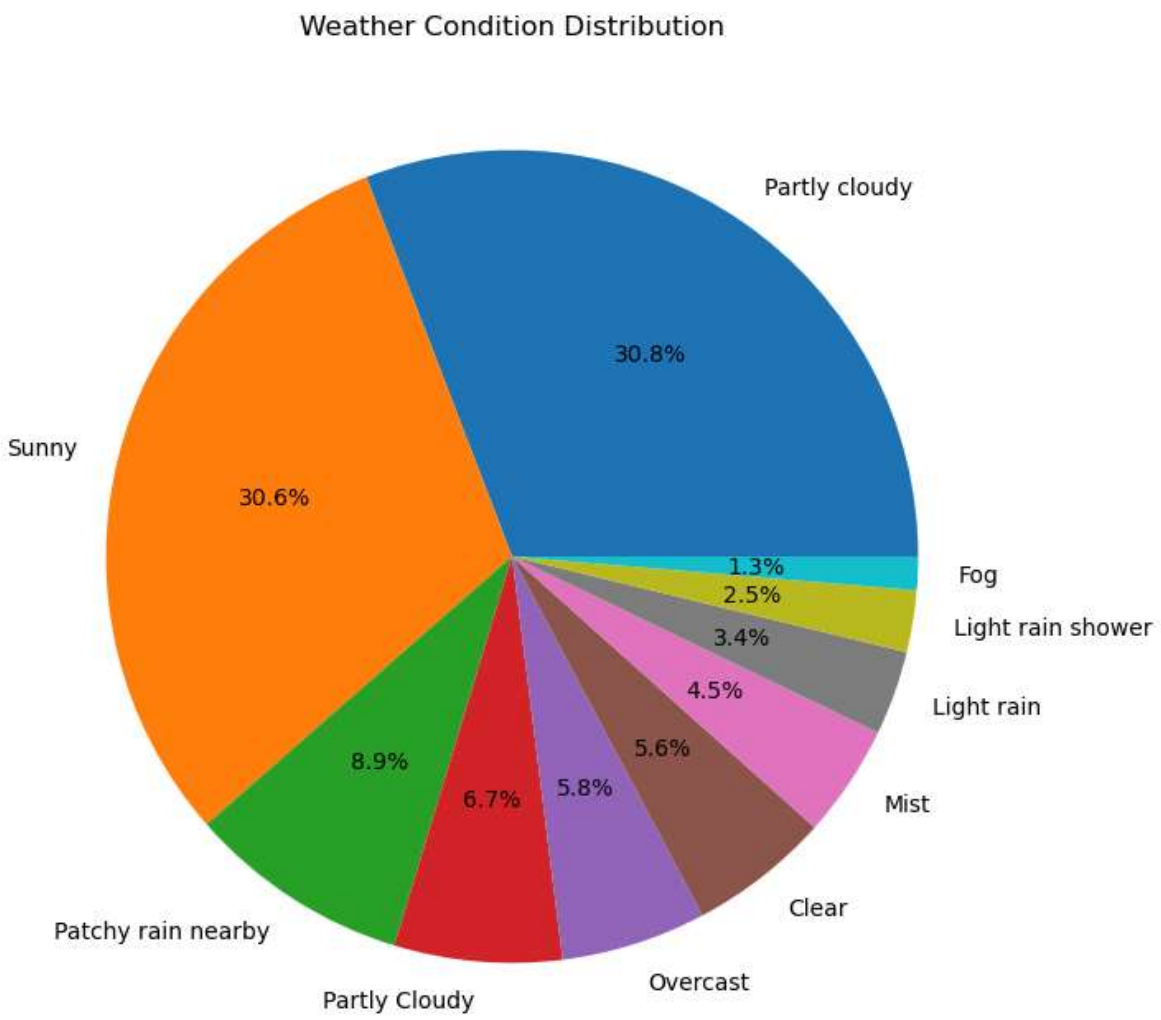


- The graph shows the relationship between temperature and humidity. As temperature increases, humidity tends to decreases.It helps to observe patterns or trends, such as whether humidity increases or decreases with temperature.

Weather Condition Count

- The graph shows the most common weather condition.
- A few conditions have high count, while others are less common.

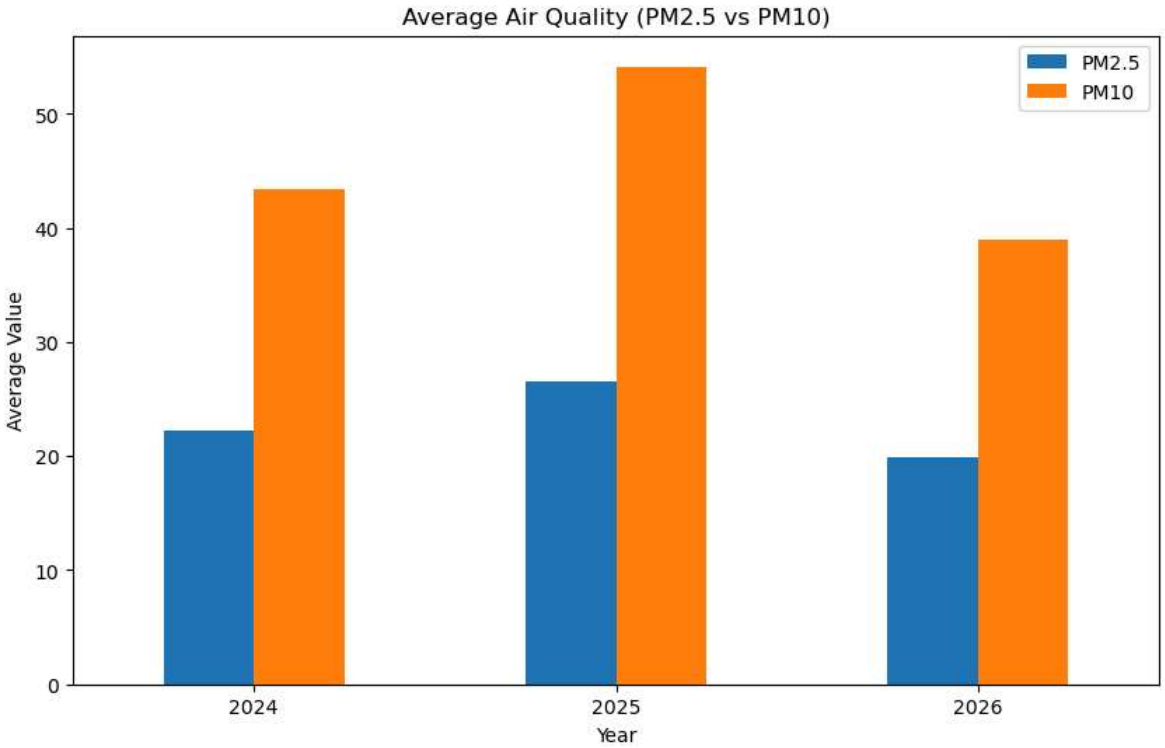
In [9]:
```
data = df['condition_text'].value_counts().head(10)

plt.figure(figsize=(8,8))

plt.pie(
    data.values,
    labels=data.index,
    autopct='%1.1f%%'
)

plt.title('Weather Condition Distribution')

plt.show()
```



- This pie chart shows the proportion of different weather conditions, helping to understand which conditions occur most frequently.

Average Air Quality

In [10]:
```
year_avg = df.groupby("year")[["air_quality_PM2.5", "air_quality_PM10"]].mean()


year_avg.plot(kind="bar", figsize=(10,6))

plt.title("Average Air Quality (PM2.5 vs PM10)")
plt.xlabel("Year")
plt.ylabel("Average Value")

plt.xticks(rotation=0)
plt.legend(["PM2.5", "PM10"])

plt.show()
```

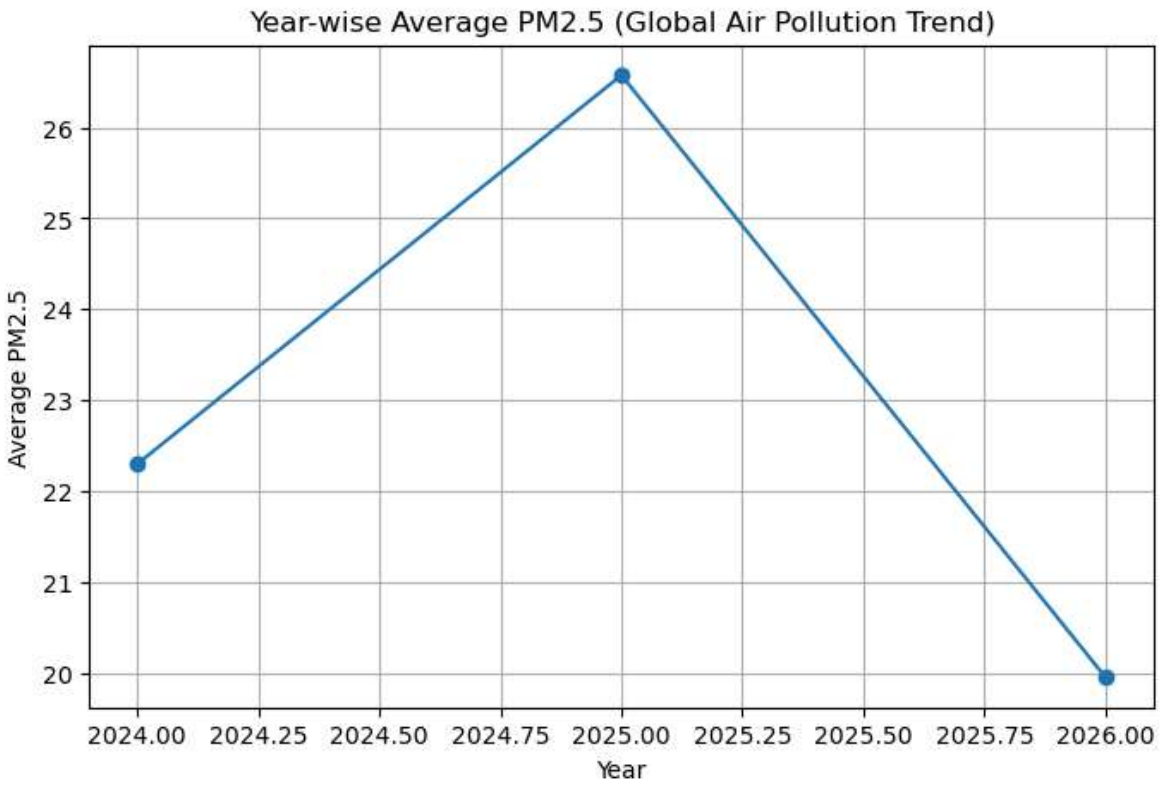


- This graph shows how average PM2.5 and PM10 levels change across different years, helping to compare air pollution trends over time.

25/04/2026 Day 6

Global Air Pollution

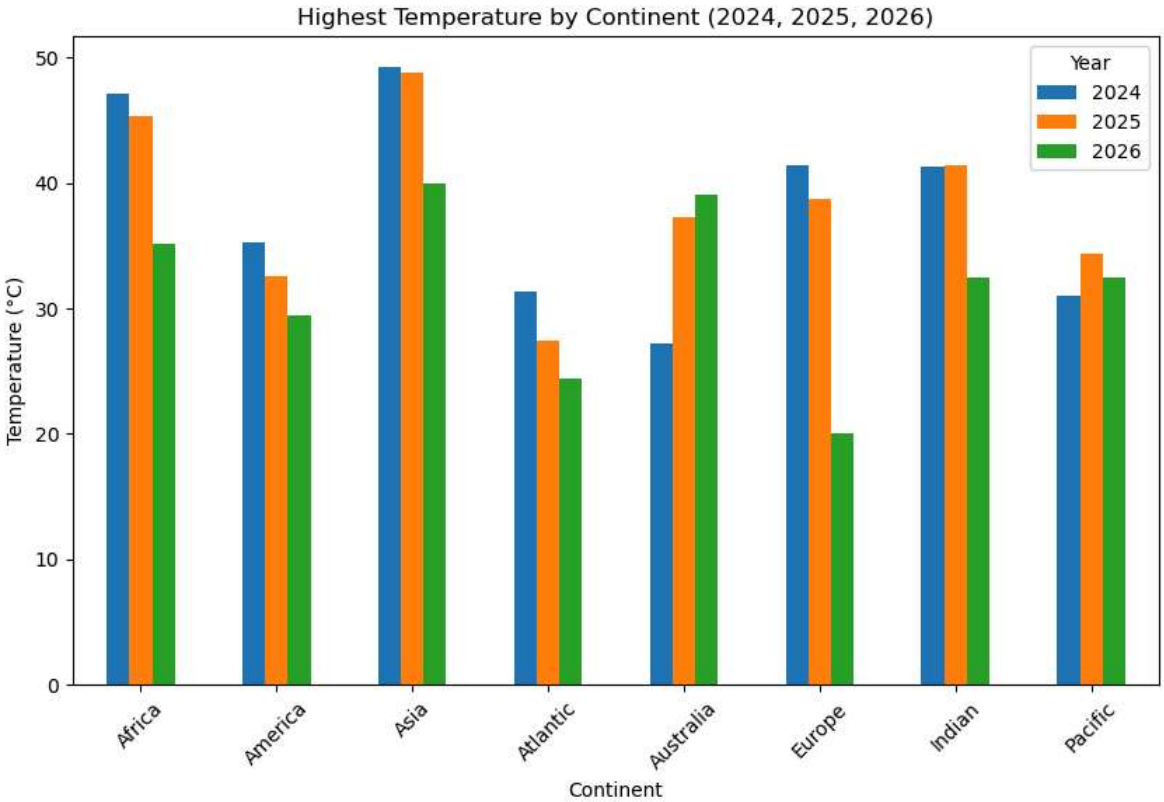
In [11]:
```
# Convert date column to datetime
df['last_updated'] = pd.to_datetime(df['last_updated'])

# Extract year
df['year'] = df['last_updated'].dt.year

# Group by year and get average PM2.5
yearly_data = df.groupby('year')['air_quality_PM2.5'].mean()

# Line chart
plt.figure(figsize=(8,5))
plt.plot(yearly_data.index, yearly_data.values, marker='o')

plt.title("Year-wise Average PM2.5 (Global Air Pollution Trend)")
plt.xlabel("Year")
plt.ylabel("Average PM2.5")

plt.grid(True)
plt.show()
```



- The line chart shows how average PM2.5 levels change over different years. An upward trend indicates increasing air pollution, while a downward trend shows improvement in air quality.

Highest Temperature by continent

In [12]:
```
df_filtered = df[df["year"].isin([2024, 2025, 2026])]

# Group data
data = df_filtered.groupby(["continent", "year"])["temperature_celsius"].max().u

# Plot
data.plot(kind="bar", figsize=(10,6))

plt.title("Highest Temperature by Continent (2024, 2025, 2026)")
plt.xlabel("Continent")
plt.ylabel("Temperature (°C)")

plt.xticks(rotation=45)
plt.legend(title="Year")

plt.show()
```

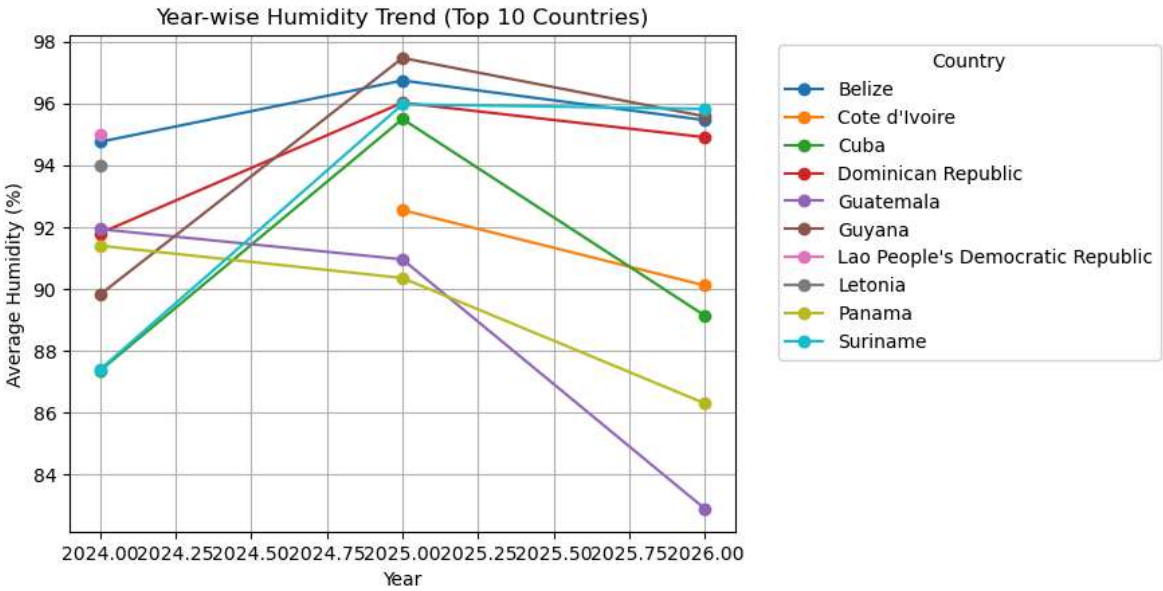


- The graph shows the **highest temperature recorded in each continent**. Some continents have **very high temperatures**, indicating hotter climate regions.

Top 10 Countries by Average Humidity

In [13]:
```
# Convert date column
df['last_updated'] = pd.to_datetime(df['last_updated'])

# Extract year
df['year'] = df['last_updated'].dt.year

# Get top 10 countries by overall humidity
top10 = df.groupby("country")["humidity"].mean().sort_values(ascending=False).he

# Filter only those countries
filtered_df = df[df['country'].isin(top10)]

# Group by year and country
yearly_data = filtered_df.groupby(['year', 'country'])['humidity'].mean().unstac

# Line chart
plt.figure(figsize=(20,10))
yearly_data.plot(marker='o')

plt.title("Year-wise Humidity Trend (Top 10 Countries)")
plt.xlabel("Year")
plt.ylabel("Average Humidity (%)")

plt.grid(True)
plt.legend(title="Country", bbox_to_anchor=(1.05,1))
plt.show()
```

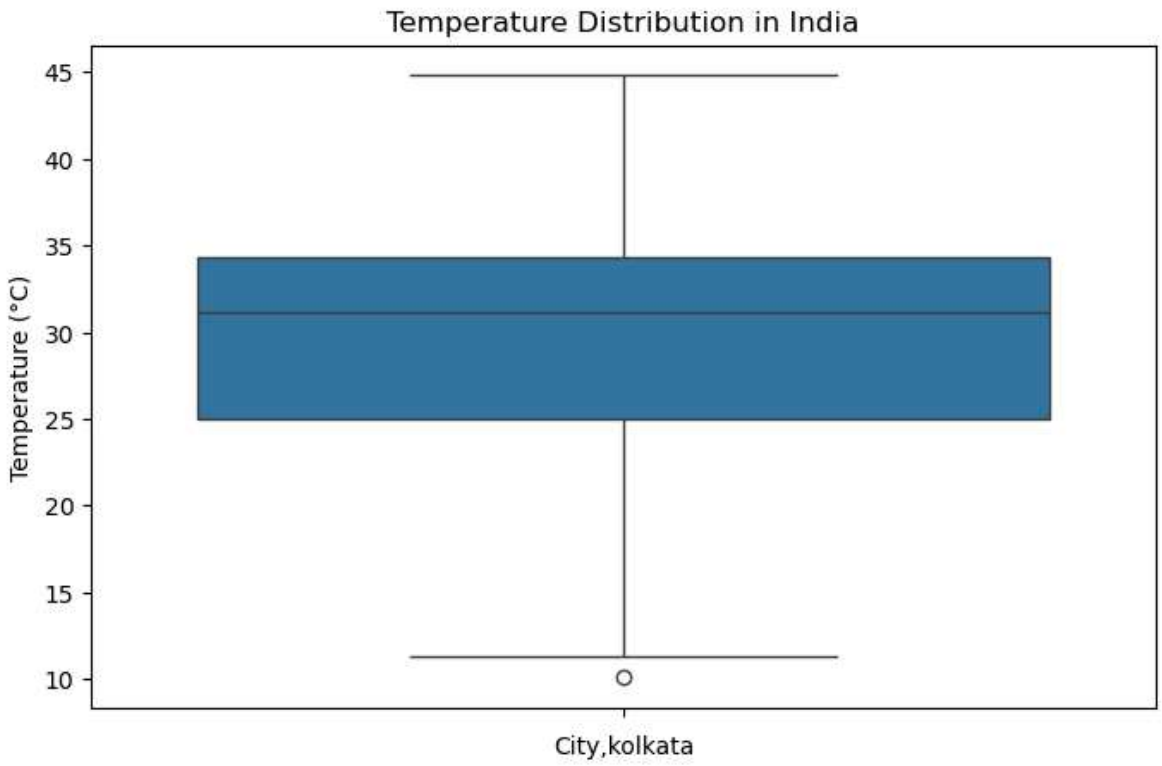
```
<Figure size 2000x1000 with 0 Axes>
```



- This chart shows how humidity changes each year in top 10 countries. If the line goes up → humidity increases, if it goes down → humidity decreases.

Temperature Distribution in India

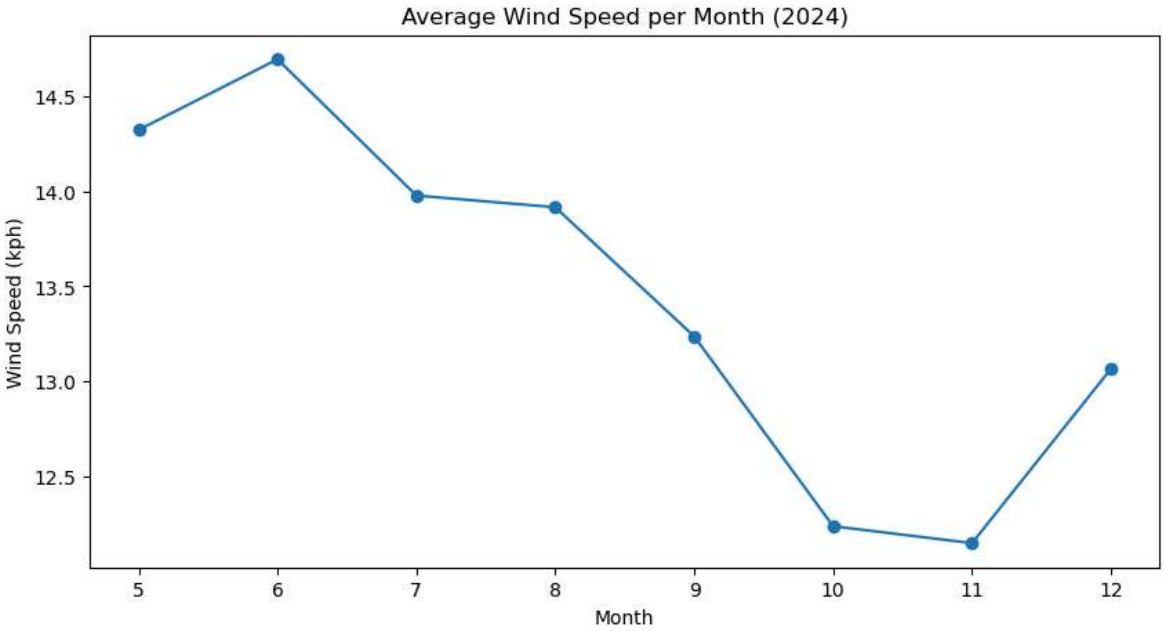
In [14]:
```
# Clean column names
df.columns = df.columns.str.strip()

# Filter only India
india_df = df[df["country"] == "India"]

# Box plot
plt.figure(figsize=(8,5))

sns.boxplot(
    data=india_df,
    y="temperature_celsius"
)

plt.title("Temperature Distribution in India")
plt.xlabel("City,kolkata")
plt.ylabel("Temperature (°C)")

plt.show()
```



- This box plot shows the distribution of temperature in India, highlighting the median, variation, and extreme values (outliers). It helps understand overall temperature patterns

Average Wind Speed Per Month

In [15]:
```
df.columns = df.columns.str.strip()

# Filter 2024
df_2024 = df[df["year"] == 2024]

# Monthly average
monthly_avg = df_2024.groupby("month")["wind_kph"].mean()

# Plot
plt.figure(figsize=(10,5))
monthly_avg.plot(marker='o')

plt.title("Average Wind Speed per Month (2024)")
plt.xlabel("Month")
plt.ylabel("Wind Speed (kph)")

plt.show()
```



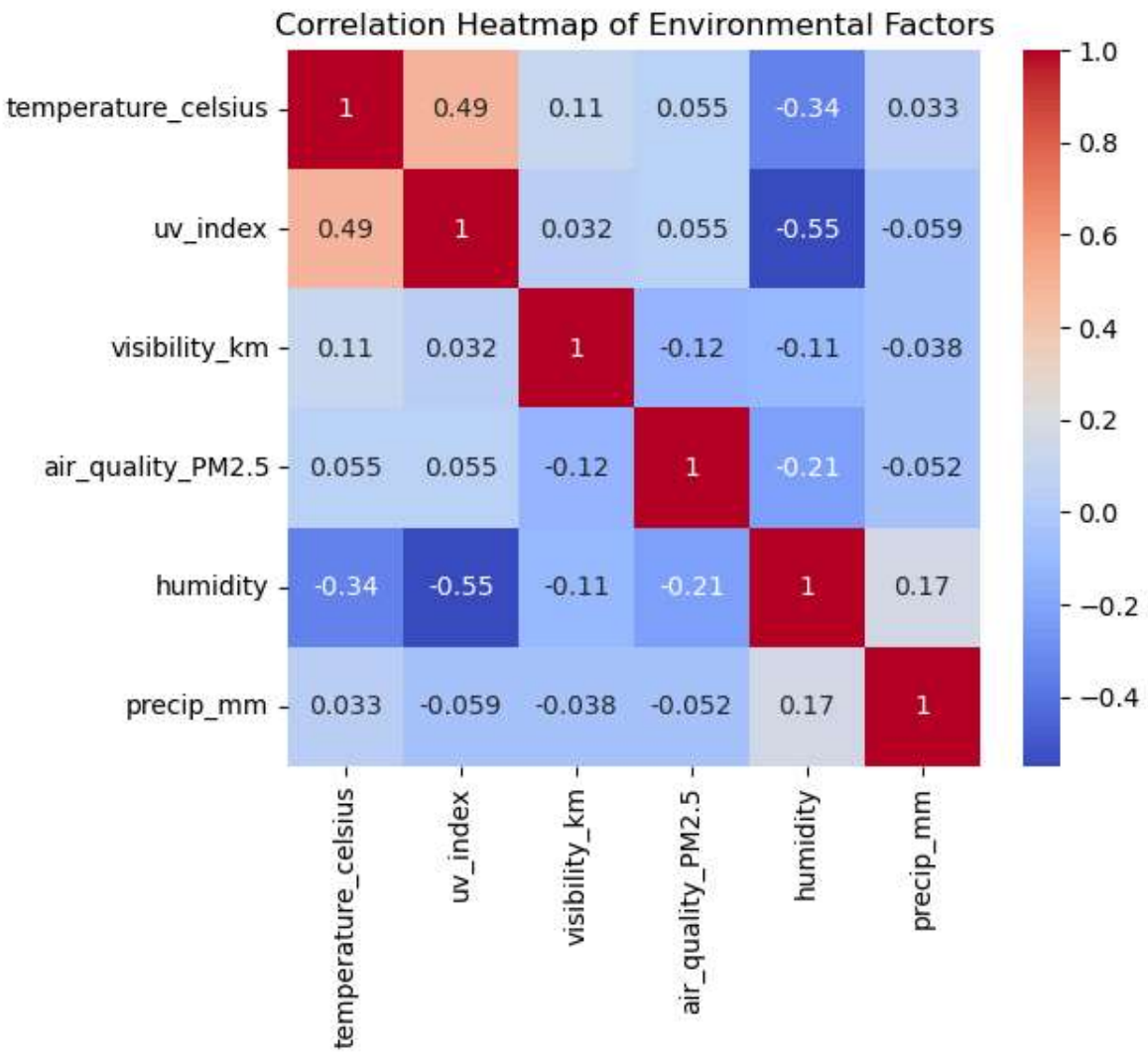
- **Wind speed in 2024 varies across months, showing seasonal changes. Some months have higher speeds, with a few extreme values indicating strong winds.**

Day 7

CORRELATION GRAPH

Correlation Heatmap of Environmental Factors

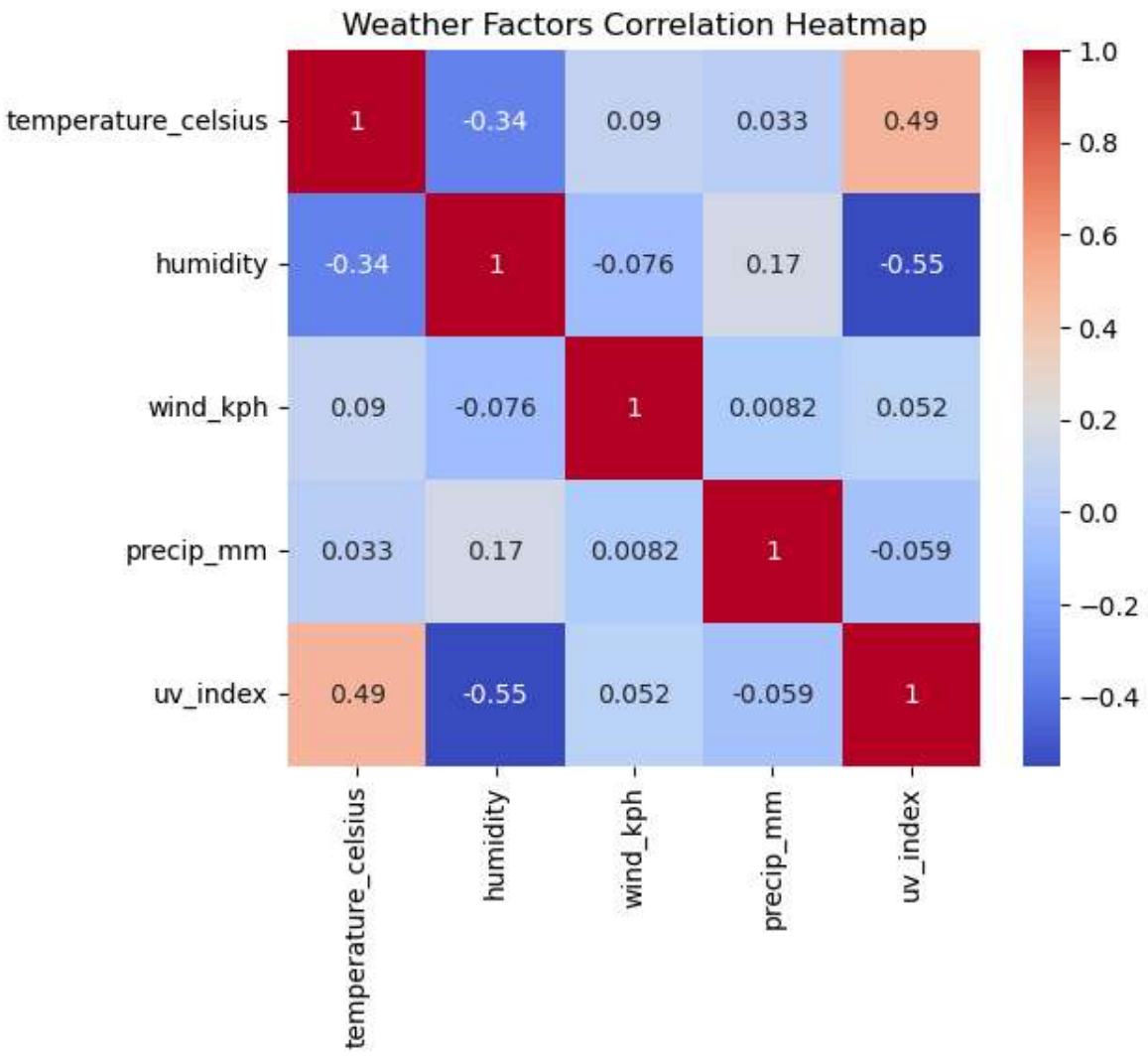
In [16]:
```
selected = df[[
    'temperature_celsius',
    'uv_index',
    'visibility_km',
    'air_quality_PM2.5',
    'humidity',
    'precip_mm'
]]

corr = selected.corr()

plt.figure(figsize=(6,5))
sns.heatmap(corr, annot=True, cmap='coolwarm')

plt.title('Correlation Heatmap of Environmental Factors')
plt.show()
```



- This heatmap shows how environmental factors like temperature, humidity, UV index, rainfall, and air pollution are related. Values show whether the factors have a strong positive, negative, or weak relationship.

Weather Factor Correlation Heatmap

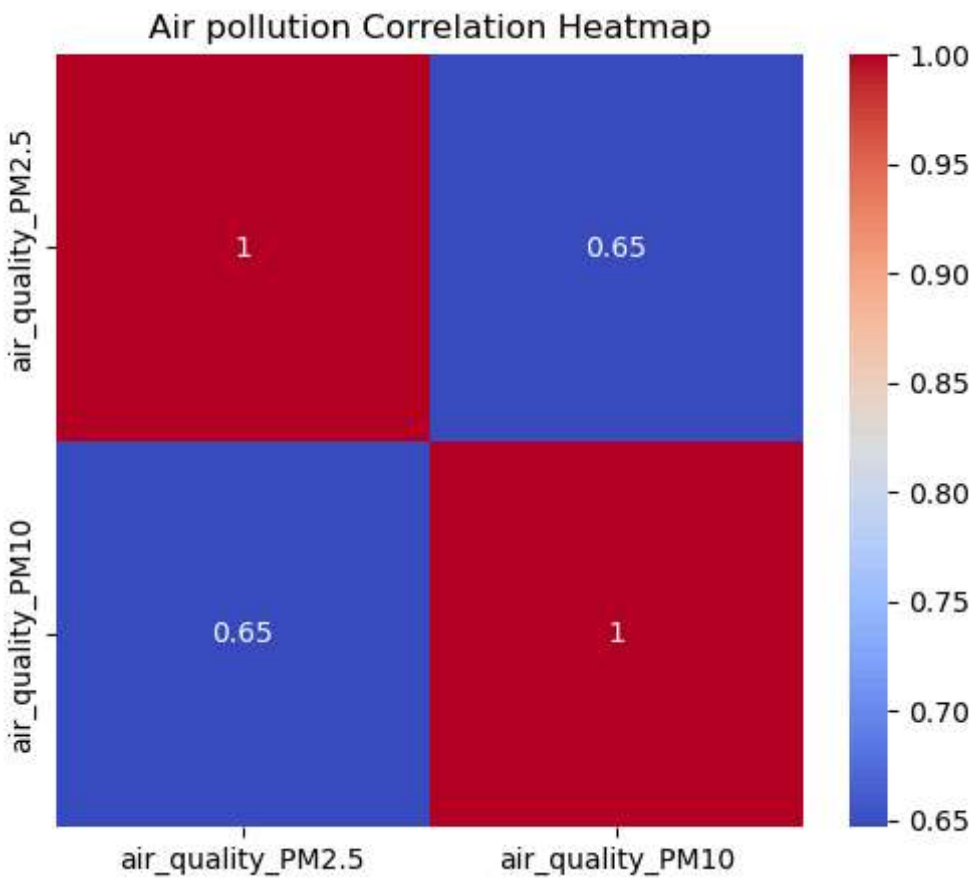
In [17]:
```
weather = df[[
    'temperature_celsius',
    'humidity',
    'wind_kph',
    'precip_mm',
    'uv_index'
]]

corr = weather.corr()

plt.figure(figsize=(6,5))
sns.heatmap(corr, annot=True, cmap='coolwarm')

plt.title("Weather Factors Correlation Heatmap")
plt.show()
```



- Temperature increases while humidity decreases, and it is positively related to the UV index.
- Humidity is negatively related to the UV index, meaning higher humidity gives lower UV values.
- Wind speed and precipitation show very weak relationships with the other factors.

Air Pollution Correlation heatmap

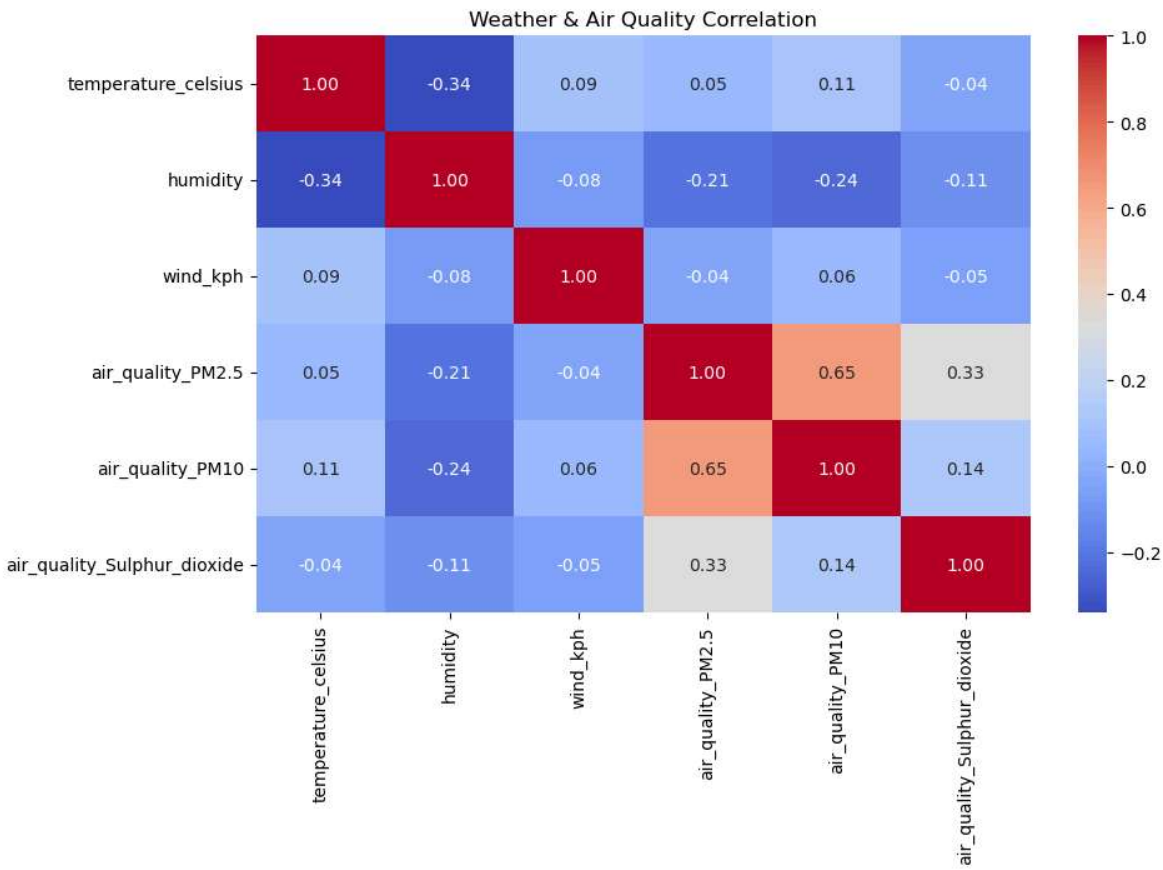
In [18]:
```
selected = df[['air_quality_PM2.5','air_quality_PM10']]

corr = selected.corr()

plt.figure(figsize=(6,5))
sns.heatmap(corr, annot=True, cmap='coolwarm')

plt.title('Air pollution Correlation Heatmap')
plt.show()
```



- PM2.5 and PM10 show a strong positive correlation, meaning both increase or decrease together. This indicates they come from similar pollution sources like dust, smoke, and vehicle emissions.

Weather & Air Quality Correlation

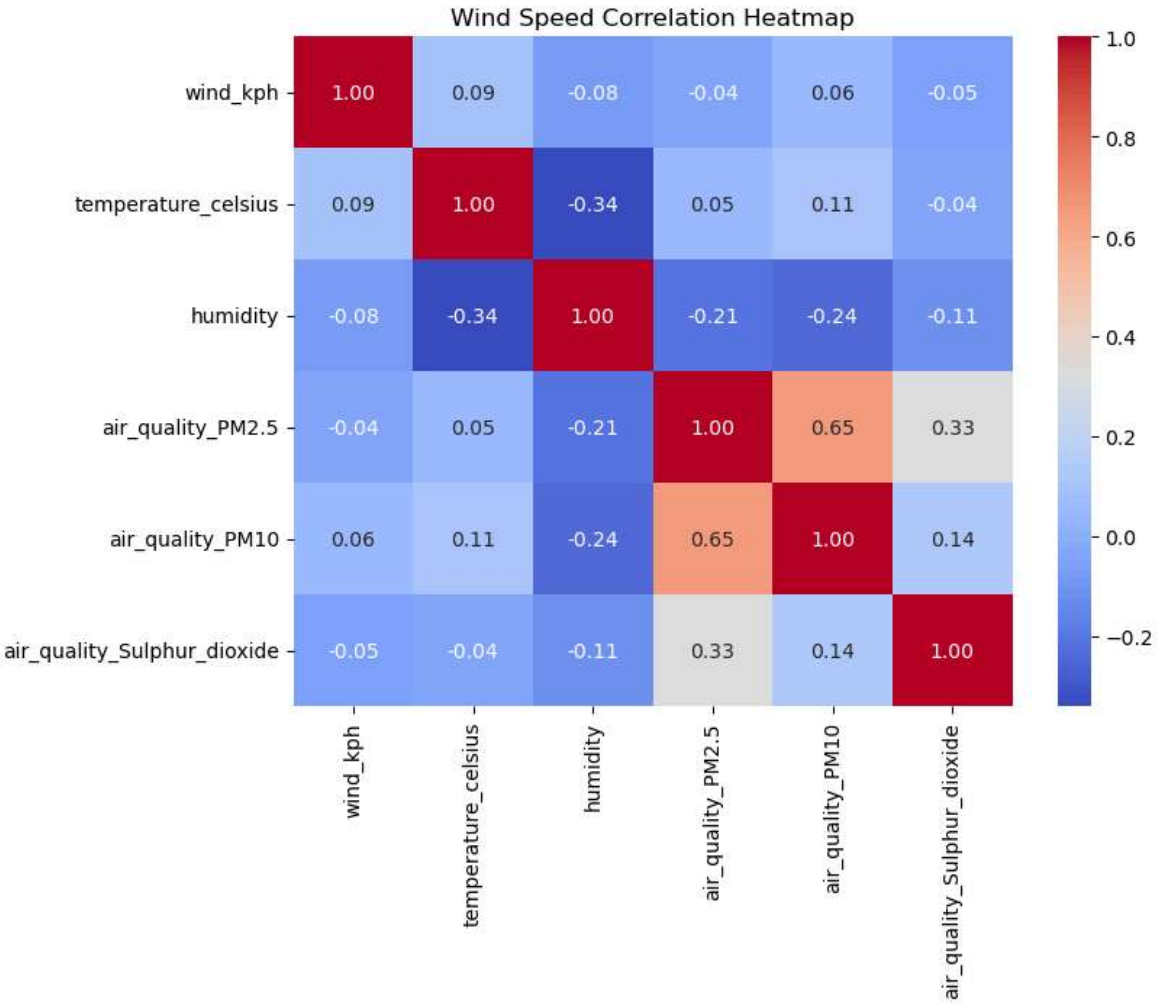
In [19]:
```
cols = [
    "temperature_celsius",
    "humidity",
    "wind_kph",
    "air_quality_PM2.5",
    "air_quality_PM10",
    "air_quality_Sulphur_dioxide"
]

corr = df[cols].corr()

plt.figure(figsize=(10,6))
sns.heatmap(corr, annot=True, cmap="coolwarm", fmt=".2f")

plt.title("Weather & Air Quality Correlation")
plt.show()
```



- Temperature, humidity, and wind show varying (mostly weak to moderate) relationships with air pollution levels. Air pollutants (PM2.5, PM10, SO_2) tend to be more closely related to each other than to weather factors.

Wind Speed Correlation Heatmap

In [20]:
```
cols = [
    "wind_kph",
    "temperature_celsius",
    "humidity",
    "air_quality_PM2.5",
    "air_quality_PM10",
    "air_quality_Sulphur_dioxide"
]

selected = df[cols]

corr = selected.corr()

import matplotlib.pyplot as plt
import seaborn as sns

plt.figure(figsize=(8,6))
sns.heatmap(corr, annot=True, cmap="coolwarm", fmt=".2f")

plt.title("Wind Speed Correlation Heatmap")
plt.show()
```



- Wind speed has weak to moderate correlation with temperature and humidity. It shows low correlation with air pollution, meaning wind does not strongly affect pollution levels directly.

In [ ]: